

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Salesforce Developer”

Submitted in partial fulfillment for the award of degree (20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Tazyeen Fathima - 1DS20AI053

Conducted at



Smartbridge

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**Salesforce Developer**” carried out by **Ms Tazyeen Fathima [1DS20AI053]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect to the course Internship (20AI8ICINT)

**Signature of Reporting
Head**

Tarakesh Kola
Salesforce Developer
SmartBridge

**Signature of Internship
Coordinator**

Prof. Kavya D N
Dept. of AI & ML
DSCE

**Signature of HoD
Dr. Vindhya P Malagi**

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Tazyeen Fathima**, final year student of Artificial Intelligence and Machine Learning Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **SmartBridge**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

Place: Bangalore

USN: 1DS20AI053

NAME: TAZYEEN FATHIMA

OFFER LETTER



Dear Learner,

Greetings from SmartInternz!

I hope this message finds you well. I am thrilled to announce that you have successfully onboarded for the upcoming **Salesforce Virtual Internship Cohort 5!**

The journey for this cohort is about to commence, and we are excited to launch it in November 2023. To guide you through this enriching experience, we have attached the **Cohort 5 Calendar and Handbook.**

The attached documents will provide you with essential information, timelines, and details about the program structure, enabling you to make the most of this internship opportunity.

Cohort 5 Calendar below: This calendar outlines important dates, sessions.
<https://drive.google.com/file/d/1--IO15Y6F7FDEMHv6nAZuh5eV8la-3Dp/view?usp=sharing>

Cohort 5 Handbook: This handbook serves as a comprehensive guide, offering insights into the program, expectations, and valuable resources.

<https://drive.google.com/file/d/18ULhkGq7eNE8Byjd3qAXaL48BcTyHzEZ/view?usp=sharing>

COMPLETION CERTIFICATE



CERTIFICATE OF COMPLETION
February 26, 2024
Tazyeen Fathima
Salesforce Developer Virtual Internship

During the 8 Weeks period of Virtual Internship (**November 2023 - January 2024**), Tazyeen Fathima has completed the following Salesforce Trailhead modules

Salesforce Fundamentals
Organizational Setup
Relationship & Process Automation
Types Of Flows & Security
Apex, Testing & Debugging
VS Code Setup & CLI Setup
Lightning Web Components (LWC) & API

Super Badge - Apex Specialist
Super Badge - Process Automation Specialist
Super Badge - Developer Super Set

Certificate ID: SISFVIPAD2024-93458 | Verify this certificate @ https://smartinternz.com/internships/salesforce_certificates/6215ff596443b9c761b344f67c85eaaad


Shri Buddha Chandraseker
Chief Coordinating Officer(CCO),
NEAT Cell-AICTE


Mr Amarender Katkam
Founder & CEO, SmartBridge &
SmartInternz

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out the Internship and for her valuable guidance and support.

We would like to thank Mr Tarakesh Kola, Smartbridge, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of the Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not least, we would like to thank our parents and friends without whose constant help, the completion of the Internship would have not been possible.

TAZYEEEN FATHIMA [1DS20AI053]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1
2	DESCRIPTION OF ORGANIZATION	2-4
3	SCOPE OF WORK	5-6
4	LEARNING OBJECTIVES AS GIVEN BY THE COMPANY	7-10
5	CHALLENGES AND SOLUTIONS	11
6	ACHIEVEMENTS AND CONTRIBUTION	12-13
7	REFLECTION AND ANALYSIS	14
8	RECOMMENDATIONS	15
	CONCLUSION	16
	REFERENCES	17

CHAPTER 1

INTRODUCTION

The Salesforce Developer Virtual Internship Program for 2024 represents a cutting-edge opportunity for aspiring developers to immerse themselves in the dynamic world of Salesforce development. This program is meticulously crafted to offer participants a comprehensive learning experience, blending theoretical knowledge with hands-on practice to cultivate a robust skill set in Salesforce development. From building custom applications to designing intuitive user interfaces, interns will explore a range of development concepts tailored to real-world scenarios.

One of the hallmark features of this internship program is its emphasis on experiential learning. Interns will have the chance to apply their newfound knowledge through hands-on projects, working on real-life development tasks under the guidance of experienced mentors. This practical experience not only reinforces theoretical concepts but also instills confidence and proficiency in applying them to solve practical challenges.

Moreover, the program fosters a collaborative learning environment, where interns can engage with peers, exchange ideas, and tackle complex problems together. Through group discussions, code reviews, and collaborative projects, participants will sharpen their communication and teamwork skills, essential for success in today's tech-driven workplace.

In addition to technical skills development, the internship program also places a strong emphasis on professional growth and career readiness. Interns will have access to resources and guidance to enhance their resume, refine their interview skills, and navigate the job market effectively. Furthermore, networking opportunities with industry professionals and alumni will provide valuable insights and connections for future career prospects.

Overall, the Salesforce Developer Virtual Internship Program for 2024 offers a transformative learning experience, empowering participants to become proficient Salesforce developers ready to make a meaningful impact in the ever-evolving tech industry. With its blend of theoretical knowledge, hands-on experience, and professional development support, this program equips interns with the skills and confidence to thrive in the competitive landscape of Salesforce development.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Salesforce is a leading cloud-based customer relationship management (CRM) platform that empowers businesses to connect with their customers in innovative ways. With a wide range of features and capabilities, Salesforce enables organizations to manage customer interactions, streamline sales processes, automate tasks, and gain valuable insights into customer behavior. It offers customizable solutions for sales, marketing, service, commerce, and more, catering to diverse industries and business needs. Salesforce's intuitive interface, scalability, and a robust ecosystem of third-party integrations make it a preferred choice for businesses of all sizes seeking to enhance customer engagement and drive growth.

Smartbridge is a technology consulting firm specializing in helping companies leverage the power of digital transformation to drive business success. With expertise in a variety of industries, including oil and gas, manufacturing, healthcare, and retail, Smartbridge offers strategic guidance and implementation services to help organizations harness emerging technologies for competitive advantage. Whether it's implementing enterprise resource planning (ERP) systems, developing custom software solutions, or optimizing business processes, Smartbridge collaborates closely with clients to deliver tailored solutions that align with their unique goals and challenges. Through a combination of technical expertise, industry insights, and a client-centric approach, Smartbridge empowers businesses to thrive in the digital age.

CHAPTER 3

SCOPE OF WORK

The scope of work in Salesforce is vast and multifaceted, encompassing a wide range of activities related to the implementation, customization, administration, and maintenance of Salesforce solutions. Some key areas within the scope of Salesforce work include:

1. **Implementation and Configuration:** This involves setting up and configuring Salesforce to meet the specific needs and workflows of an organization. It includes tasks such as defining data models, creating custom objects, fields, and relationships, configuring user permissions and security settings, and setting up automation rules.
2. **Custom Development:** Salesforce provides a robust development platform, allowing developers to build custom applications, integrations, and functionalities on top of the Salesforce platform using languages like Apex (similar to Java) and Visualforce (a markup language). Custom development may involve building custom business logic, triggers, workflows, and user interfaces tailored to the unique requirements of the organization.
3. **Integration:** Salesforce often needs to be integrated with other systems and applications within an organization's IT landscape, such as ERP systems, marketing automation platforms, or external databases. Integration work involves designing and implementing data integration solutions, API integrations, and middleware configurations to ensure seamless data flow and synchronization between systems.
4. **Data Management:** Salesforce administrators are responsible for managing data within the Salesforce platform, including data import/export, data cleansing, deduplication, and data quality maintenance. This may involve using Salesforce data tools such as Data Loader, Data Import Wizard, or third-party data management solutions.
5. **Administration and Support:** Salesforce administrators play a crucial role in managing and maintaining the health of the Salesforce instance. This includes tasks such as user management, permission sets, role hierarchy, profiles, and troubleshooting issues related to system performance, security, or data integrity. Administrators also handle ongoing user support and training.

6. Reporting and Analytics: Salesforce offers robust reporting and analytics capabilities to help organizations gain insights into their sales, marketing, and service operations. Salesforce professionals may be involved in creating custom reports and dashboards, defining key performance indicators (KPIs), and analyzing data to make informed business decisions.

7. Continuous Improvement and Innovation: Salesforce professionals are tasked with staying abreast of new features, updates, and best practices within the Salesforce ecosystem. They actively seek opportunities to optimize existing processes, improve user adoption, and drive innovation by leveraging new Salesforce technologies and functionalities.

Overall, the scope of work in Salesforce is dynamic and evolving, offering a diverse range of opportunities for professionals with expertise in Salesforce administration, development, integration, and analytics.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Learning objectives in a Salesforce Developer Virtual Internship Program, as outlined by Smartbridge and SmartInternz, may include:

1. **Fundamentals of Salesforce Development:** Gain a deep understanding of the Salesforce platform, including its architecture, data model, and core functionalities.
2. **Apex Programming:** Learn the Apex programming language, which is used for developing custom business logic and integrations within the Salesforce platform.
3. **Visualforce and Lightning Component Development:** Master the development of user interfaces using Visualforce and Lightning Web Components to create rich, interactive experiences for Salesforce users.
4. **Data Modeling and Management:** Understand data modeling concepts in Salesforce and learn best practices for designing efficient data models, creating custom objects, and managing data relationships.
5. **Integration Techniques:** Explore various integration techniques to connect Salesforce with external systems and applications, including REST and SOAP APIs, middleware platforms, and third-party integration tools.
6. **Salesforce Automation:** Learn how to automate business processes using Salesforce automation tools such as workflows, process builder, and flow builder to streamline operations and improve efficiency.
7. **Testing and Debugging:** Develop skills in testing and debugging Salesforce applications to ensure quality and reliability, including writing unit tests, debugging Apex code, and using Salesforce development tools for code analysis.
8. **Version Control and Deployment:** Understand version control concepts and learn how to use tools like Git to manage Salesforce code repositories. Gain hands-on experience with Salesforce deployment tools for packaging and deploying changes across environments.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. **Technical Complexity:** Interns may find aspects of Salesforce development, such as Apex programming or Lightning component development, challenging due to their technical complexity.

Solution: Provide comprehensive training materials, including tutorials, documentation, and hands-on exercises, to help interns build a strong foundation in Salesforce development concepts. Offer mentorship and support from experienced developers to guide interns through challenging topics and answer questions.

2. **Communication Barriers:** Remote communication and collaboration can sometimes be challenging, leading to misunderstandings or delays in project progress.

Solution: Foster clear and effective communication channels, such as regular video calls, instant messaging platforms, and virtual collaboration tools. Provide guidelines for professional communication and encourage interns to proactively reach out to mentors or peers for assistance when needed.

3. **Time Management:** Interns may struggle to balance their internship responsibilities with other commitments or distractions while working remotely.

Solution: Set clear expectations regarding internship hours, deadlines, and deliverables. Encourage interns to create structured schedules, allocate dedicated time for internship tasks, and minimize distractions during working hours. Offer time management tips and techniques to help interns optimize their productivity.

4. **Technical Support and Resources:** Interns may encounter technical issues or lack access to necessary resources and support while working remotely.

Solution: Provide comprehensive technical support, including access to online forums, helpdesk support, and troubleshooting guides. Ensure that interns have access to all required software tools, development environments, and documentation. Offer virtual office hours or support sessions where interns can seek assistance from mentors or technical experts.

5. Engagement and Motivation: Maintaining intern engagement and motivation in a virtual setting can be challenging due to the lack of face-to-face interaction and social connection.

Solution: Implement strategies to foster a sense of community and belonging, such as virtual team-building activities, social networking groups, and online events. Recognize and celebrate interns' achievements and milestones to boost morale and motivation. Encourage regular feedback and communication to keep interns engaged and motivated throughout the program.

By proactively addressing these challenges and implementing effective solutions, a Salesforce Developer Virtual Internship Program can provide a rewarding learning experience for participants, equipping them with valuable skills and preparing them for success in the field of Salesforce development.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Individual Achievements:

a. **Mastery of Salesforce Technologies:** During the internship specific Salesforce technologies and tools were mastered, such as Apex programming, Lightning components, Visualforce pages, Salesforce administration and development skills, etc.,

b. **Successful Project Deliveries:** A few projects were completed independently and also as part of a team with the successful implementation of Salesforce solutions, meeting project milestones and deadlines, and delivering high-quality results.

c. **Certifications Obtained:** Post internship Salesforce developer certifications were obtained such as Salesforce Certified Developer, Salesforce Certified Administrator, or specialized certifications in areas like Sales Cloud or Service Cloud.

2. Contributions to Team/Organization:

a. **Enhancements to Salesforce System:** Based on insights gained during the internship, enhancements were made to the organization's Salesforce system such as optimizing workflows, implementing automation, or improving data management processes.

b. **Technical Guidance and Support:** Technical guidance or support was provided such as assisting with troubleshooting issues, sharing best practices, or conducting training sessions.

c. **Collaborative Projects:** Collaborative projects were undertaken during the internship, driving to successful outcomes, fostering teamwork, and contributing to the overall success of the project.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship at Salesforce through Smartbridge was an enriching journey that provided me with a multifaceted learning experience. Working extensively with MuleSoft integration, I gained a deep understanding of APIs, integration patterns, and the intricacies of the Anypoint Platform. From designing to deploying APIs and managing integrations, I was immersed in the technical intricacies that underpin modern digital connectivity. This hands-on experience not only solidified my technical proficiency but also equipped me with the skills necessary to tackle real-world integration challenges effectively.

Collaborating within Smartbridge teams and engaging directly with clients was a pivotal aspect of my internship experience. Through these interactions, I honed my project management and communication skills, learning how to navigate complex project scopes, gather comprehensive requirements, and present tailored solutions to meet clients' specific needs. Engaging in client-facing roles allowed me to understand the importance of effective communication and building strong relationships in delivering successful projects. Moreover, troubleshooting issues in real-time provided me with invaluable experience in agile problem-solving, a skillset essential for any professional endeavor.

Being part of the Salesforce ecosystem provided me with unique insights into the broader industry landscape. Understanding how MuleSoft integrates seamlessly with Salesforce CRM and Marketing Cloud expanded my perspective on the potential of integration technologies in driving business innovation and efficiency. Additionally, networking opportunities within the Salesforce and MuleSoft communities allowed me to forge connections with industry professionals and gain exposure to emerging trends and best practices. Overall, my internship at Salesforce through Smartbridge not only deepened my technical expertise but also equipped me with invaluable soft skills and industry insights that will undoubtedly shape my future career trajectory.

CHAPTER 8

RECOMMENDATION

In conclusion, this virtual internship has significantly enhanced my understanding and skills in Salesforce development. The hands-on tasks and challenges presented throughout the program were instrumental in bridging the gap between theoretical knowledge and practical application. Engaging directly with real-world scenarios has provided me with a solid foundation in Salesforce, allowing me to navigate and solve complex problems efficiently.

The mentorship aspect of the internship was particularly beneficial. The mentors provided personalized feedback and guidance that was crucial to my learning process. Their expertise in Salesforce development not only deepened my knowledge but also inspired confidence in my abilities as a developer. The constructive critiques and one-on-one sessions helped refine my techniques and strategies in handling various Salesforce projects.

Overall, the internship offers a comprehensive and engaging learning experience that is both challenging and rewarding. It is an excellent opportunity for anyone aspiring to become a proficient Salesforce developer. The structured yet flexible curriculum is designed to cater to both beginners and those with some experience, ensuring that all participants gain a competitive edge in the field. I highly recommend this program to any individual seeking professional growth and expertise in Salesforce development.

CHAPTER 9

CONCLUSION

My internship experience at SmartBridge has not only provided me with technical skills but has also instilled in me a strong sense of adaptability and resilience in dynamic work environments. Throughout various challenges and opportunities encountered during the internship, I have consistently demonstrated the ability to learn quickly and thrive in fast-paced settings. This adaptability, coupled with my continuous drive for improvement, positions me as a candidate capable of tackling new challenges with enthusiasm and determination.

Looking ahead, I am committed to furthering my professional growth and making meaningful contributions to the organizations and communities I engage with. Whether it's through continued learning, mentorship, or active participation in industry events, I am dedicated to staying at the forefront of technological advancements and best practices. My internship experience has not only been a stepping stone in my career but has also laid a solid foundation for future success.

In summary, my internship journey at SmartBridge has been a transformative experience that has shaped me into a well-rounded professional ready to take on the challenges of tomorrow. With a strong foundation in technical expertise, project management skills, and a client-centric approach, I am poised to make significant contributions in any role I undertake. I am grateful for the opportunities afforded to me during this internship and excited about the possibilities that lie ahead as I continue to grow and evolve in my career.

CHAPTER 10

REFERENCES

- [1] https://github.com/28042002/Salesforce_virtual_internship
- [2] <https://www.thesmartbridge.com/>
- [3] <https://www.salesforce.com/in/>
- [4] <https://www.trailhead.com/>
- [5] <https://trailhead.salesforce.com/users/smartbridgesupport/trailmixes/developerhttps://smartinternz.com/>
- [6] https://smartinternz.com/Student/guided_projects